

Transcriptional subtypes of hepatocellular carcinoma and their predicted genetic dependencies

TCGA-LIHC NMF subtyping → TCGADEPMAP dependency mapping

Abstract

We defined transcriptional subtypes of hepatocellular carcinoma (HCC) by applying non-negative matrix factorization (NMF) to TCGA-LIHC bulk RNA-seq [1], then projected the published TCGADEPMAP predicted gene-essentiality map [2] onto those subtypes to identify subtype-selective dependencies. Three stable transcriptional programs emerged—S1 stromal/ECM-mesenchymal, S2 well-differentiated hepatocyte/metabolic, and S3 oncofetal-proliferative. Two correspond quantitatively to the tumor-cell axes of the established Hoshida classification [3] (S2 → Hoshida S3 differentiation, 88% of samples; S3 → Hoshida S2 proliferation), whereas S1 is a stromal/microenvironment axis orthogonal to Hoshida’s tumor-intrinsic classes ($\chi^2 p = 1 \times 10^{-21}$). Testing 1,966 predicted dependencies (ANOVA + Tukey HSD, then subtype-vs-rest Welch *t*-tests with Benjamini–Hochberg FDR) recovered 3,241 significant subtype × gene associations (FDR < 0.05). The hepatocyte-metabolic subtype S2 carried the largest set of selective dependencies, including the clinically actionable targets CDK6 and MDM2. Orthogonal validation against measured CRISPR essentiality in 22 HCC cell lines (DepMap 24Q4) directionally confirmed the S1 redox/BET pair (TXNRD1, BRD2) and the S2 cell-cycle axis (CDK6, CCND1), while MDM2 and ADAR did not validate; the aggregate signal was supportive but modest, reflecting both the biology and the small cell-line panel.

1 Methods

The full workflow—data sources, automated processing steps, and the two human-directed decision points (NMF rank selection; the choice to use precomputed rather than retrained essentiality models)—is shown in Figure 1.

1.1 Data sources

Expression. TCGA-LIHC RNA-seq was obtained from the UCSC Xena TCGA Hub (HiSeqV2: RSEM pancan-normalized, $\log_2(\text{norm_count} + 1)$, HGNC symbols). After removing normal and recurrent samples and 377 zero-variance genes, 371 primary tumors × 20,153 genes remained; the top 5,000 genes by median absolute deviation (MAD) were used as NMF input.

Predicted dependencies. The TCGADEPMAP predicted gene-essentiality matrix [2] applies expression-only elastic-net models—trained on DepMap CRISPR CERES gene-effect scores [4, 5] and transferred onto TCGA transcriptomes after quantile normalization and contrastive-PCA alignment—to predict essentiality for 1,966 cross-validated genes across 9,805 TCGA samples. The premise that expression alone captures most dependency signal is supported independently: essentiality is better predicted by mRNA abundance than by gene-regulatory-network-inferred activity [6]. More negative scores denote stronger predicted dependency. TCGA-LIHC samples were matched against the GDC TCGA-LIHC case list (377 cases), yielding 370 LIHC samples.

Analysis pipeline: automated processing and human-directed decision points

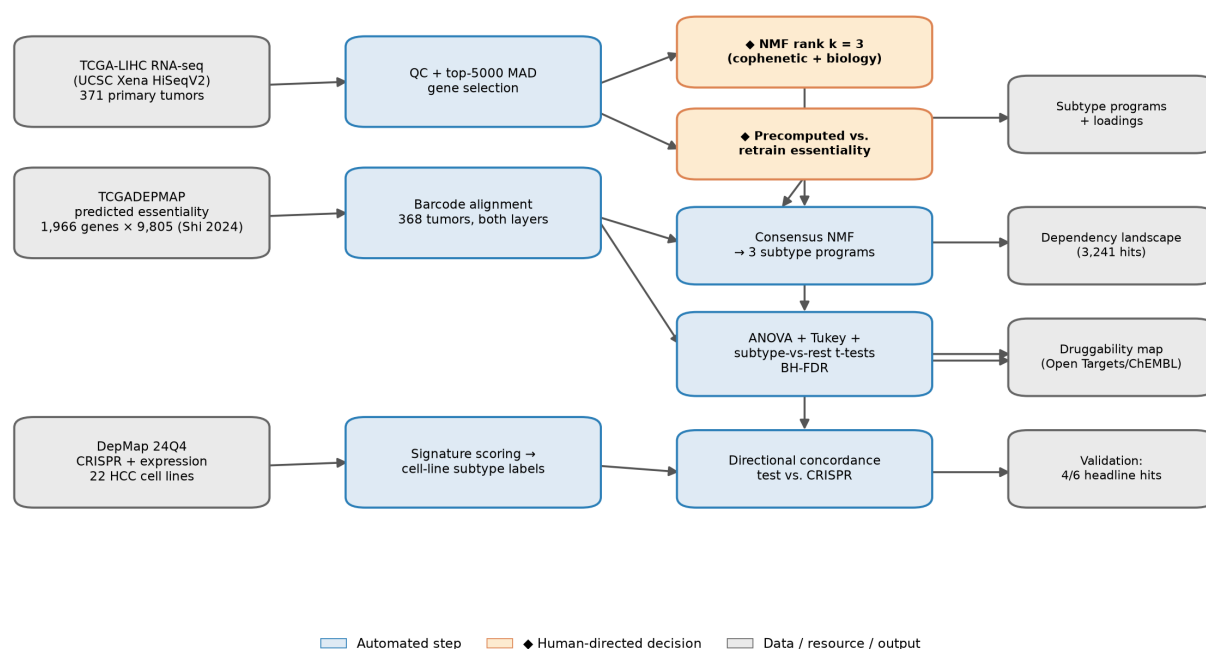


Figure 1: Analysis pipeline. Grey = data resources and outputs; blue = automated processing and statistical steps; amber diamonds = human-directed decisions. Three public resources (TCGA-LIHC expression, the TCGADEPMAP predicted-essentiality map, and DepMap 24Q4 CRISPR data for orthogonal validation) feed a consensus-NMF subtyping step, a three-substep significance-testing core, druggability annotation, and a cell-line validation leg.

CRISPR validation data. DepMap 24Q4 Public [7] CERES gene-effect and expression matrices were used for orthogonal validation; 25 HCC cell lines were identified, 22 with both data layers.

1.2 NMF subtype definition

Decomposing bulk tumor transcriptomes into a small number of additive programs and reading cellular state from the resulting factors follows the signature-decomposition paradigm [8]. NMF (scikit-learn) was run on the $371 \times 5,000$ log-expression matrix across ranks $k = 2-8$. Rank was selected by consensus clustering over 40 random restarts per rank (Brunet criterion, `init='random'`): the cophenetic correlation coefficient peaked at $k = 3$ (0.989) and declined at $k = 4$ (0.974) and $k = 5$ (0.940), while reconstruction error fell smoothly with no sharp elbow (explained variance 0.926 at $k = 3$; Figure 2). The trivial $k = 2$ split had marginally higher consensus silhouette but collapses the known three-class HCC structure, so $k = 3$ was chosen. The final model (best of 100 seeded restarts, `init='nndsvda'`, minimum reconstruction error) yielded mixture coefficients $\mathbf{W}$ (371×3) and gene-program loadings $\mathbf{H}$ ($3 \times 5,000$); each tumor was assigned to the program with maximal $\mathbf{W}$ coefficient. Both NMF stages use fixed random seeds and are deterministic.

1.3 Dependency statistics

Subtype labels were inner-joined to the essentiality matrix on 15-character barcodes, giving 368 tumors with both layers (S1=142, S2=88, S3=138). Three tests were applied: (1) per-gene one-way ANOVA across subtypes with Tukey HSD and a one-vs-rest bipartition sweep; (2) per subtype $\times$ gene, a Welch t -test of that subtype’s essentiality vs. all others pooled, with Benjamini–Hochberg FDR across all 5,898 tests, where $\Delta = \text{mean}(\text{subtype}) - \text{mean}(\text{others})$ and $\Delta < 0$ denotes stronger dependency in that subtype; (3) z -scored per-subtype mean CERES for visualization. Druggability of the top selective dependencies was annotated via the Open Targets Platform and ChEMBL. All statistics were recomputed independently in an audit (see Data/Code Availability).

1.4 Orthogonal validation

Each DepMap HCC cell line was assigned an NMF subtype label by scoring its expression on the same S1/S2/S3 program signatures. Predicted selectivity was tested two ways: a categorical comparison of measured CERES between matching-subtype lines and the rest, and a continuous Spearman correlation between a line’s own-program score and measured essentiality. Aggregate concordance was assessed by a one-sided binomial test (directional alternative > 0.5).

1.5 Hoshida correspondence

Tumors were scored on curated marker sets for the three Hoshida subclasses and cross-tabulated against the NMF assignment (χ^2 test, Cramér’s V); marker-set enrichment against NMF loadings used Fisher’s exact test.

2 Results

2.1 Three transcriptional programs

NMF resolved three reproducible programs (cophenetic correlation 0.989 at $k = 3$; Figure 2). Program identity was read from the top-loading genes and Reactome enrichment: S1 (stromal/ECM-mesenchymal; LUM, DCN, COL1A1, POSTN), S2 (well-differentiated hepatocyte/metabolic; HP,

APOA2, ADH1B, ALDOB, CPS1), and S3 (oncofetal-proliferative; AFP, GPC3, H19, PEG10, MDK). Subtype sizes were S1=142, S2=88, S3=141 ($n = 371$).

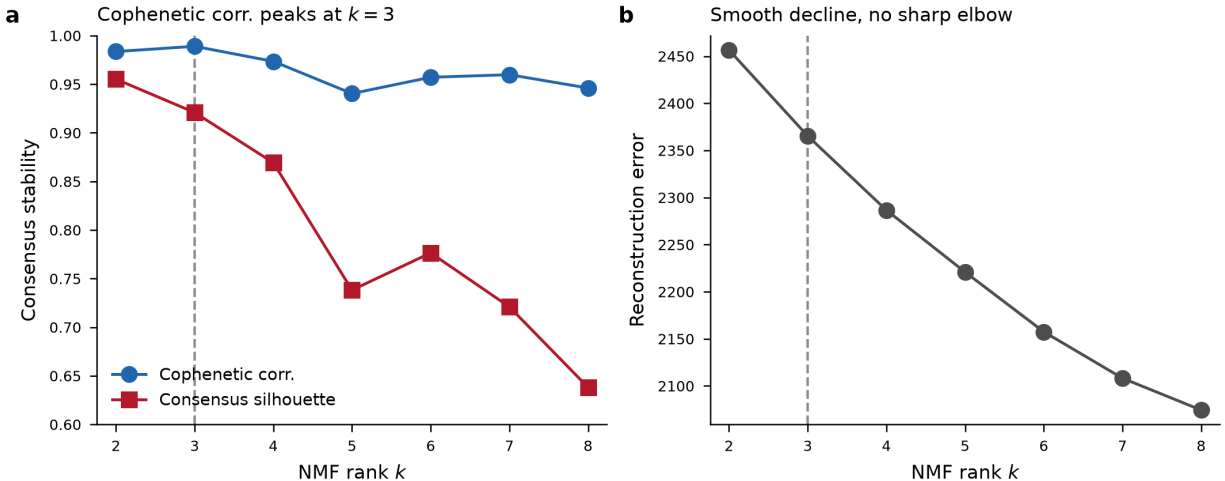


Figure 2: NMF rank selection. (a) Consensus cophenetic correlation peaks at $k = 3$; consensus silhouette generally declines over non-trivial ranks (minor uptick at $k = 6$). (b) Reconstruction error declines smoothly with no sharp elbow. $k = 3$ is the most stable non-trivial solution.

2.2 Quantified correspondence to the Hoshida classes

Rather than assert the resemblance, we tested it. Per-tumor Hoshida signature calls cross-tabulated against the NMF assignment gave $\chi^2 p = 1.0 \times 10^{-21}$ (Cramér’s $V = 0.376$; Table 1). The correspondence is strong and specific for the two tumor-intrinsic programs—NMF S2 maps to Hoshida S3 (well-differentiated; 88% of samples) and NMF S3 to Hoshida S2 (proliferative; the plurality call). Marker-set enrichment agrees (NMF-S2 vs. Hoshida-S3 differentiation markers 8/8, $p = 6 \times 10^{-12}$; NMF-S3 vs. Hoshida-S2 proliferation markers 7/12, $p = 1 \times 10^{-7}$). NMF S1, the stromal/ECM program, does not map cleanly to any single Hoshida class—its tumors distribute across all three—because the Hoshida scheme is tumor-cell-intrinsic whereas S1 is a microenvironment/purity axis. The two schemes are thus complementary rather than redundant. (These are curated literature marker sets scored by signature averaging, not the official nearest-template-prediction classifier, so this is a quantified correspondence, not a formal re-labeling.)

Table 1: NMF subtype vs. per-tumor Hoshida signature call (row-normalized %, $n = 371$).

NMF subtype	→ Hoshida S1	→ Hoshida S2 (prolif.)	→ Hoshida S3 (diff.)
S2 (hepatocyte-metabolic)	12	0	88
S3 (oncofetal-proliferative)	25	50	25
S1 (stromal/ECM)	33	27	39

2.3 Subtype-selective dependency landscape

Across 5,898 subtype $\times$ gene Welch t -tests, 3,241 were significant at $FDR < 0.05$; one-way ANOVA flagged 1,569 of 1,966 genes. Selective dependencies ($\Delta < 0$, significant) numbered 380 (S1), 800

(S2), and 437 (S3). The z -normalized dependency landscape (Figure 3) separates into three diagonal blocks. S2, the hepatocyte-metabolic subtype, carried the strongest and most numerous selective dependencies, including CDK6 ($\Delta = -0.27$, $q = 6.2 \times 10^{-16}$), MDM2 ($\Delta = -0.27$, $q = 7.7 \times 10^{-11}$), and ADAR ($\Delta = -0.39$, $q = 4.5 \times 10^{-17}$).

2.4 Druggable dependencies

Annotating the top 30 selective dependencies per subtype against Open Targets and ChEMBL, 10 of 90 targets have a target-linked drug, concentrated in S2 (7; vs. S1=2, S3=1; Figure 4). The headline actionable hits are:

- **S2 — CDK6** ($\Delta = -0.27$), targeted by approved CDK4/6 inhibitors (palbociclib, ribociclib, abemaciclib, trilaciclib); its cyclin partner **CCND1** ($\Delta = -0.25$) is co-selective (an indirect/complex-level link). CDK4 inhibition synergizes with mutant-p53 targeting in HCC [9], and cyclin D1 turnover is a documented cell-cycle control point [10]. That a well-differentiated, better-prognosis subtype is the one predicted to depend on the cell cycle is notable and would need direct validation.
- **S2 — MDM2** ($\Delta = -0.27$), targeted by Phase 2–3 MDM2–p53 antagonists (idasanutlin, navtemadlin, siremadlin, alrizomadlin). Mechanistically coherent with S2’s more wild-type-like phenotype—but this is one of the two headline predictions that did *not* validate in cell lines (Section 2.5), so it is the lowest-confidence S2 hypothesis pending direct testing.
- **S1 — TXNRD1** ($\Delta = -0.17$), an approved-drug redox target (auranofin, arsenic trioxide); TXNRD1 inhibition induces oxidative stress and apoptosis in HCC [11, 12] and TXNRD1 is upregulated to support hepatoma growth [13]. **BRD2** ($\Delta = -0.13$), a BET-family bromodomain target with documented, context-dependent activity in HCC models [14, 15].
- **S3** is the unmet-chemical-matter subtype—its leading dependencies (CHD8, RCL1, LRP-PRC, RNF168, EIF5A) are largely undrugged.

The context in which a dependency is selective is what converts a generic hit list into subtype-directed hypotheses [16]. These are drug–target links, not HCC-approved indications; the metformin/ataluren links to complex I subunits and RPL30 are indirect and should not be read as selective single-gene inhibitors.

2.5 Orthogonal validation against measured CRISPR essentiality

The predictions were tested against direct CRISPR knockout data (DepMap 24Q4; 22 HCC lines with both CRISPR and expression). Subtype labels assigned by signature scoring placed the classically mesenchymal lines (SNU449, SNU387, SNU475, HLF, SK-HEP-1) in S1 and the well-differentiated lines (HUH7, HUH1, JHH5, JHH7) in S2, consistent with their known biology (Figure 5). The validation is partial and honest:

- **Concordant.** The S1 redox/BET pair—TXNRD1 (measured $\Delta = -0.35$, $\rho = -0.32$) and BRD2 ($\Delta = -0.20$, $\rho = -0.53$)—and the S2 cell-cycle axis—CCND1 ($\Delta = -0.39$, $\rho = -0.53$) and CDK6 ($\Delta = -0.10$, $\rho = -0.34$)—are more essential in their predicted subtype, matching the tumor-level prediction.
- **Non-validating.** MDM2 ($\Delta = +0.06$) and ADAR ($\Delta = +0.07$) do not show the predicted S2-selectivity—a caution flag on two S2 headline calls, MDM2 especially.

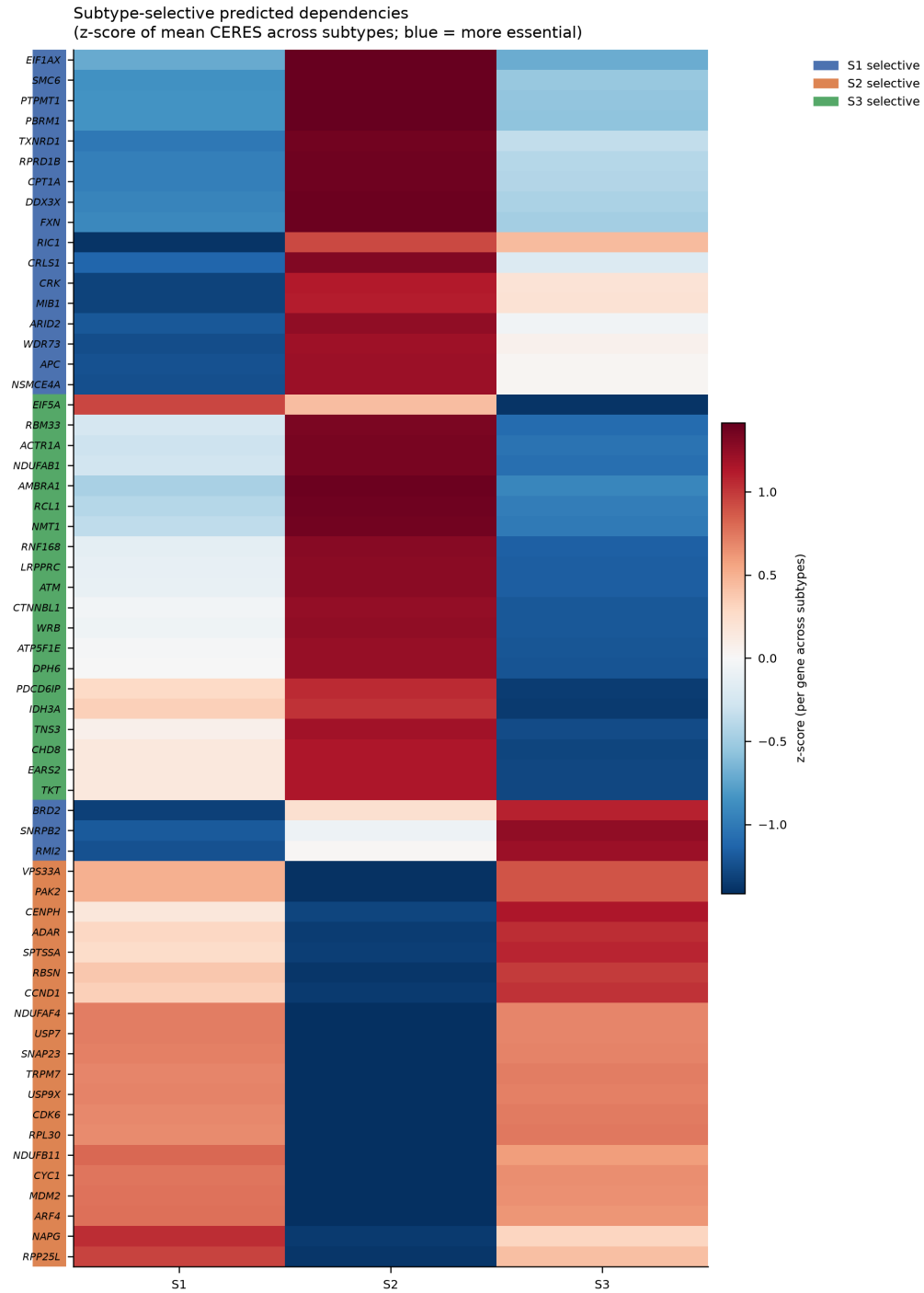


Figure 3: Subtype-selective predicted dependencies. z -score of per-subtype mean CERES (blue = more essential). The union of the top-selective genes per subtype forms a block-diagonal structure; the left colour strip marks each gene's owner subtype.

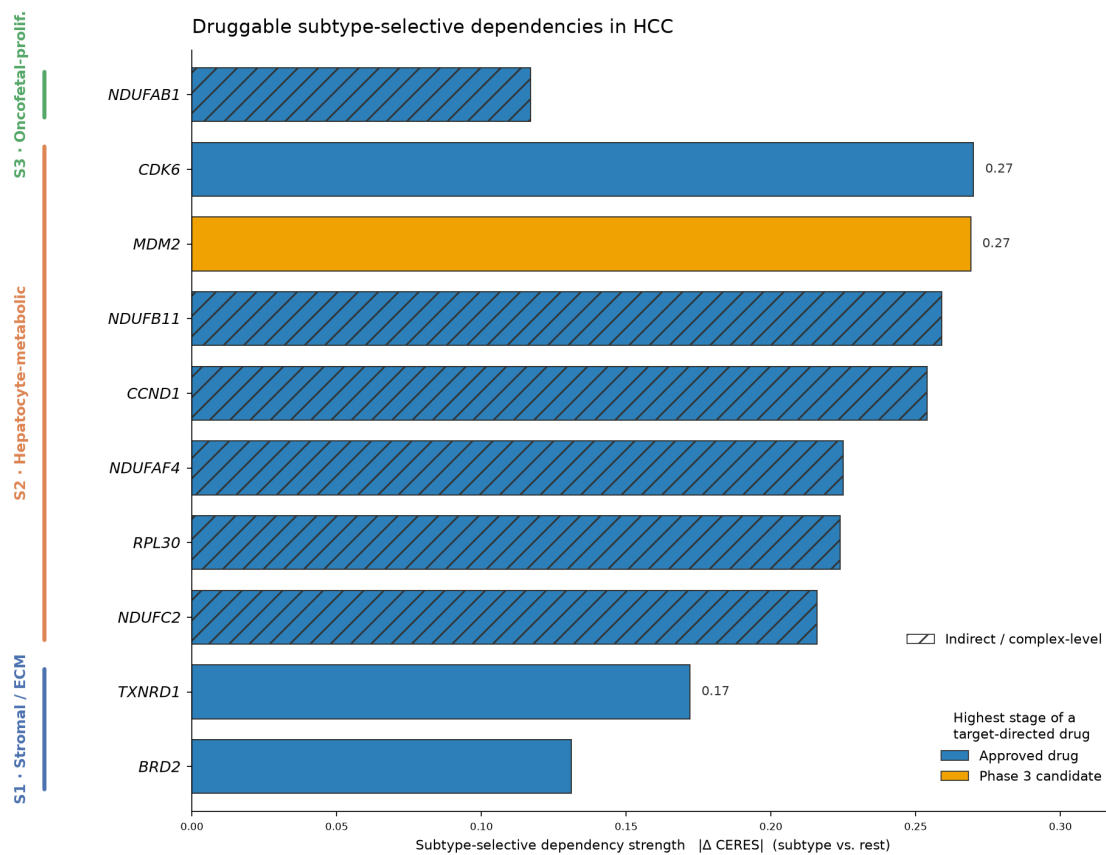


Figure 4: Druggable subtype-selective dependencies in HCC. Bar length = dependency strength $|\Delta \text{CERES}|$ (subtype vs. rest); colour = highest clinical stage of a target-directed drug (blue = approved, amber = Phase 3); hatched bars = indirect/complex-level links; left brackets group by subtype.

- **Aggregate.** Across 44 top hits, the continuous test (Spearman $\rho < 0$ in the predicted direction) is concordant for 28/44 (64%), one-sided binomial $p = 0.048$ (two-sided $p = 0.096$)—significant but modest. The categorical split sits at 21/44 (48%, chance), unsurprising given only 4–11 lines per subtype and effect sizes ($|\Delta| \approx 0.1$ –0.4 CERES units) near the resolution of a 22-line panel.

Subtype-selective transcriptional dependencies do carry into measured CRISPR essentiality for the cell-cycle and redox/chromatin axes, but the signal is weak at the single-prediction level and the panel is underpowered for a strict per-gene test. This validates the mechanism for the strongest axes and identifies MDM2/ADAR as predictions measured data does not yet support.

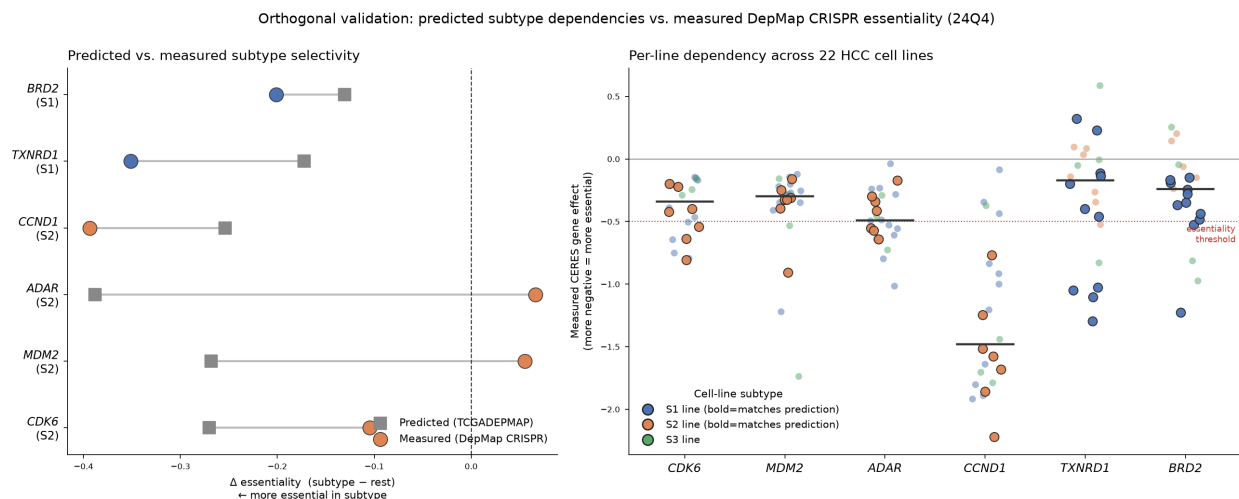


Figure 5: Orthogonal validation against DepMap 24Q4 CRISPR essentiality. (a) For six headline druggable hits, predicted subtype-selectivity (grey squares) vs. measured Δ in the matching cell-line subtype (coloured circles); points left of zero = more essential in the predicted subtype. (b) Per-line measured CERES across the 22 HCC lines; bold outlined points match each gene’s predicted subtype, black tick = median, red dotted line = the -0.5 essentiality threshold.

3 Caveats

- **Predicted, not measured, essentiality.** TCGADEPMAP scores are elastic-net predictions from expression, not direct CRISPR measurements in patient tumors, and inherit the limits of cell-line-to-tumor transfer.
- **Data-driven subtypes.** Two NMF subtypes correspond quantitatively to the Hoshida tumor-cell classes while S1 is a stromal axis; the correspondence was scored with curated marker sets, not the official classifier.
- **Predictions unsupported by cell-line data.** Validation confirmed the cell-cycle (CDK6/CCND1) and S1 redox/BET (TXNRD1/BRD2) axes but not MDM2 or ADAR; the 22-line panel is underpowered for strict per-gene validation, so a non-validating result is a caution flag rather than a refutation.
- **Stromal confounding.** The S1 program is ECM/fibroblast-dominated, so its profile—and its predicted dependencies—is influenced by non-tumor content and lower purity.

- **Drug matches are hypotheses.** Target–drug links describe drugs that act on the target, not agents approved or tested in HCC.

Data and Code Availability

All source data are public: TCGA-LIHC expression (UCSC Xena HiSeqV2), the TCGA DEPMAP predicted-essentiality map [2], and DepMap 24Q4 Public [7]. All code to reproduce every figure, table, and this manuscript—layered pipeline scripts, saved intermediate checkpoints, a pinned environment file, and an independent correctness audit—is available at <https://github.com/USERNAME/REPO>. A third party can regenerate the figures from the shipped checkpoints and compile this PDF from a clean checkout following the repository README.

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